St. Louis Local Service Directory

Relational database of St. Louis service providers, their offerings, and source evidence • Fall 2026

Scope and intended user

The database is meant for St. Louis residents and students, especially people new to the city, who want to find a local provider for a specific service and confirm where that claim came from. It covers seven categories: automotive, fitness and wellness, plumbing, healthcare, pet care, printing, and landscaping. Each business has one selected St. Louis location and at least one explicitly advertised service. All pages were collected on October 5, 2026. The sample is deliberately small and bounded. It records the services each provider clearly advertises, not its complete inventory.

Businesses	Service types	Categories	Offerings	Source pages	Publishers
15	35	7	42	21	16

Cleaning standardized city spelling and mapped equivalent wording to one controlled service name (“Oil and Filter Change” to *Oil change*, “Lab Work” to *Laboratory services*, “Group Training” to *Group exercise classes*). Each business and each business/service pair is stored once, and shared services reuse one Service row. Three facts were cross-checked against a second publisher with no conflicts. Hoffman Plumbing’s phone number matches its own site and its BBB profile. Mercy Hospital’s address matches mercy.net and the AMA FREIDA listing. Missouri Baptist Medical Center’s address matches BJC Health and CMS.

Business rules

1. **Business identity.** Each business is stored once, with a stable *business_id* and one selected location. Name, address, and city are required. A website is optional and is stored as NULL when it cannot be verified.
2. **Service classification.** Each service type belongs to exactly one category, and a category can group many services. Equivalent wording from different businesses maps to the same Service row.
3. **Offerings.** A business can offer many services and a service can be offered by many businesses. Each business/service pair is recorded at most once. Every business in the final sample has at least one recorded offering.
4. **Evidence.** Every business and every offering must reference an existing Source page and the raw observation row it came from, so any fact can be traced back to its saved capture.

Model and implementation

The schema has five tables. **Source** (PK *source_id*) stores the publisher, page title, URL, collection date, and raw-file path. **ServiceCategory** (PK *category_id*) names the seven categories. **Service** (PK *service_id*) has a required foreign key *category_id*, which models the one-to-many “Groups” relationship. **Business** (PK *business_id*) has a required foreign key *source_id* plus *raw_row_id* for its identity evidence. **BusinessService** resolves the many-to-many “Offers” relationship. Its composite primary key (*business_id*, *service_id*) blocks duplicate pairs (rule 3), and its own *source_id* and *raw_row_id* record the evidence for that offering (rule 4). NOT NULL constraints cover every column except *website* (rule 1). Foreign keys use the default restrict behavior, so referencing rows must be removed before the rows they reference. The **BusinessDirectory** view exposes the public business columns used by Q6. The “at least one offering” part of rule 3 cannot be expressed as a key, so test T1 checks it instead. Tests T2 and T3 confirm that an invalid foreign key and a duplicate pair are rejected. The ER diagram on page 3 shows every attribute, key, cardinality, and participation.

Normalization to 3NF

Start from one combined relation with a row per recorded offering: *R*(business_id, business_name, address, city, website, biz_source_id, biz_raw_row_id, service_id, service_name, category_id, category_name, offer_source_id, offer_raw_row_id, publisher, page_title, url, collection_date, raw_file_path). The candidate key is (*business_id*, *service_id*), since each business/service pair appears once.

Determinant	Dependent attributes	Problem in R
business_id	business_name, address, city, website, biz_source_id, biz_raw_row_id	Partial dependency (depends on part of the key)
service_id	service_name, category_id	Partial dependency

Determinant	Dependent attributes	Problem in R
category_id	category_name	Transitive (key → service_id → category_id)
source_id	publisher, page_title, url, collection_date, raw_file_path	Transitive through each evidence reference
business_id, service_id	offer_source_id, offer_raw_row_id	None (full key)

Removing the partial dependencies gives Business, Service, and an offering table keyed by both IDs. Removing the transitive dependencies moves category names into ServiceCategory and page metadata into Source. In each resulting table, every determinant is a key, so all five tables are in 3NF (and BCNF). This removes an **update anomaly**. In *R*, ArchWell Health's address would be repeated on all six of its offering rows, so a change could reach some rows and miss others. Now it is stored once in Business. The keys reconnect the data. BusinessService joins to Business on *business_id* and to Service on *service_id*, Service joins to ServiceCategory on *category_id*, and both Business and BusinessService join to Source on *source_id*. The foreign keys guarantee that every join finds a matching row.

Two findings from the query results

Finding 1: oil change is the most widely offered service, but not every auto shop lists it (Q2, Q4). Q2 returns three St. Louis providers with a recorded oil-change offering: St. Louis Auto & Truck Repair (business 1, raw row R001), AutoMax of St. Louis (2, R003), and Meineke Shop #30 (4, R007). Each row carries the URL of the page that supports it. Q4, the NOT EXISTS complement, returns the other 12 businesses. Car-X Tire & Auto (3) is the only automotive business among them, because its captured page supports brake repair and wheel alignment only. A resident who needs an oil change therefore has three documented options, and each can be traced to its source page.

Finding 2: shared services are rare (Q3, Q5). Q3 keeps services with two or more distinct providers and returns only six of the 35 service types: Oil change (3), Brake repair (2), Tire rotation (2), Personal training (2), Group exercise classes (2), and Laboratory services (2). A hand count from the 42 inserted offerings gives the same six groups and counts. The other 29 types each have exactly one provider. Healthcare shows this most clearly. It has 14 service types and 15 offerings, but only Laboratory services is shared (Affinia Healthcare and ArchWell Health). Q5's LEFT JOIN lists all 35 services and shows a business ID for only three of them at Fit City Gym (gym access, tanning, and sauna access). None of those three appears in Q3. Club Fitness and The L.A.B. Gym share personal training and group classes, so within fitness, Fit City has a distinct recorded profile.

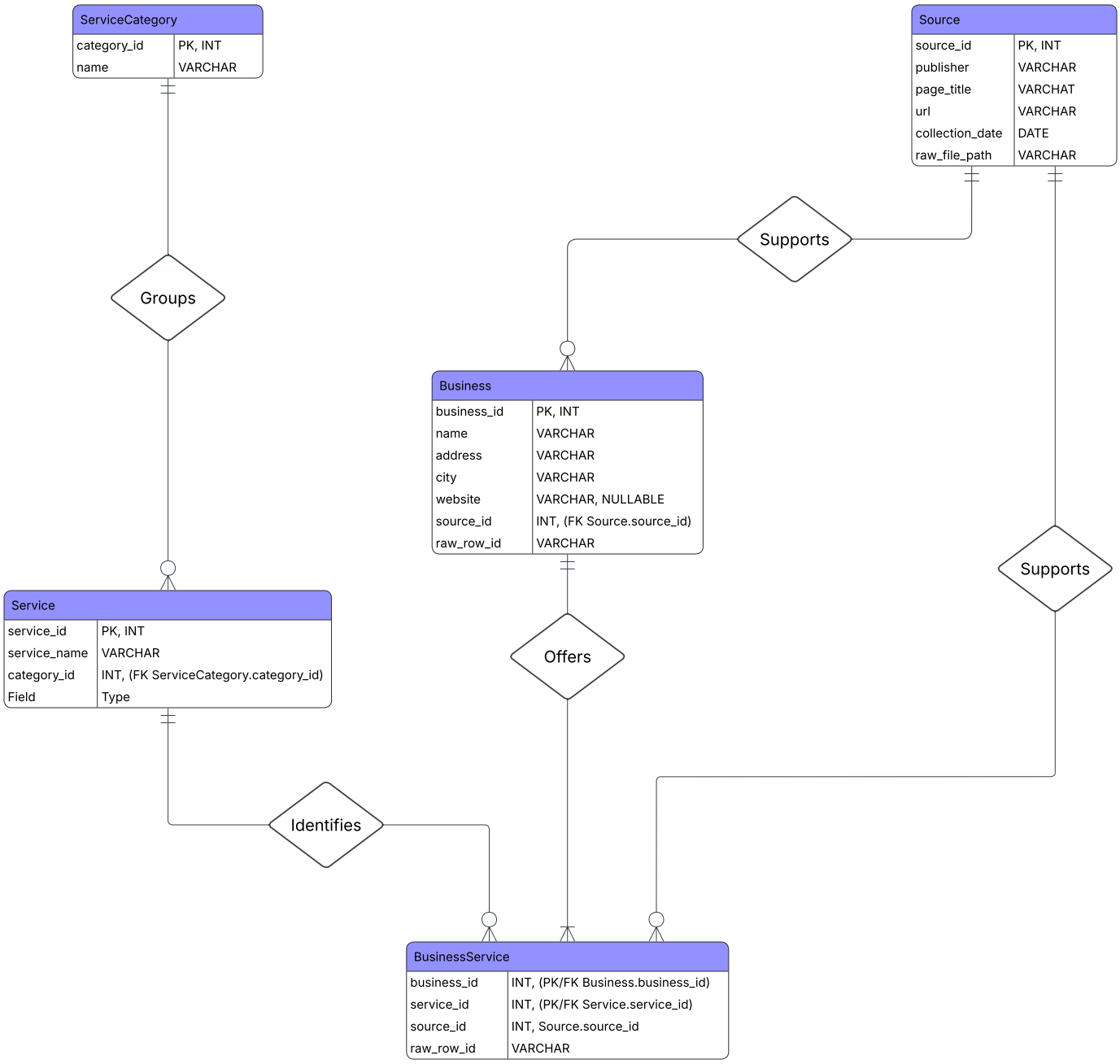
Limitation

A missing offering means "not recorded," not "not offered." The sample captures only the services each business clearly advertised on the pages we saved. Q4's list of 12 businesses without an oil change, and the 29 single-provider services in Q3, reflect how the data was collected as much as the market. For example, Car-X may well offer oil changes on a page we did not save. The same applies to Q6. It returns no rows because all 15 businesses have a verified website. The NULL-website behavior is therefore demonstrated only by the T4 test fixture, which is rolled back afterward.

Follow-up question

If we recorded the complete service list for every business, using the same controlled service names, how many of the 29 single-provider services would gain a second provider, and would oil change remain the most widely offered service?

ER diagram



Crow's-foot notation: a double bar means exactly one (mandatory), a circle with a crow's foot means zero or more (optional), and a bar with a crow's foot means one or more. Every foreign key is NOT NULL, so each child row references exactly one parent row. A Business has one or more offerings (rule 3, checked by T1). BusinessService PK = (business_id, service_id). Website is the only nullable attribute.